Original E0: constructed-kernel conformance

27 October 2026

The native query helper now constructs the full cofactor kernel for the original public E0 fixture when supplied with points from both trace cosets. This is a source-conformance result under the fixture's previously recorded ambient order. It does not supply an IC cost or an unknown logarithm.

Source: local component `44b65a769533`, linked against the verified `974f2e9cd46b` library. **Native development gate:** 15 tests pass; ordinary Cargo integration and full-library/hosted results are separate.

Fixed fixture and input control

The model is `EC1N22Cbinhd39a9df391db`, over the polynomial-basis field with modulus 4,194,307. Its equation is $y^2 + xy = x^3 + 1$. Existing metadata supplies $\#E = 4,193,912$, $r = 991$, $h = 4,232$ and $G = (97,814,2,630,359)$. This check does not repeat or replace the earlier curve-order audit.

The first positive fixture supplied independent reference lifts at $x = 0, \ldots, 63$. It generated only 2,116 kernel points, and the helper correctly refused it. The failed native attempt is preserved: build exit 0, test exit 101, 14 passes and one failure. The corrected test retains that proper span as a negative control, then adds reference lifts at $x = \text{trace_one XOR } i$, for $i = 0, \ldots, 63$. Here trace_one is the first polynomial-basis element of absolute trace one.

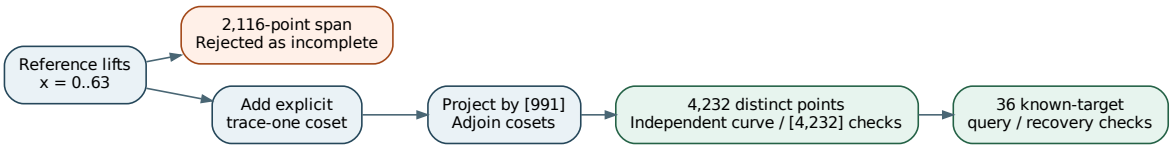


Figure 1: Deterministic supplied-point construction and independent checks. Counts are correctness checks, not runtime or yield measurements.

Verified scope

Check	Recorded outcome
Complete supplied-point span	4,232 distinct kernel points
Independent point arithmetic	Every point is on E0 and satisfies $[4,232]P = O$
Query/recovery conformance	36 fixed known-target cases agree
Combined native test executable	15 passed, zero failed
Original failed candidate set	Explicit incomplete-span control retained

Construction uses standard projection by $[r]$, successive cosets and a final census check. Candidate discovery and random sampling belong to the caller. The logs used by the recovery checks are supplied from known synthetic scalars; no factor-base relation or unknown logarithm is discovered. Setup and checks remain chargeable backend calls. Relation collection, logarithm/linear-algebra integration, sampling/yield, complete cold price and the general Lean counting bridge remain open. No crossover or whole-pipeline gain follows.

Evidence: the first attempt and corrected attempt retain commands, source/library hashes, patches and raw logs. Final build/test exits are 0/0. This report supplies no new timing result.